Chapter – 2
Swiggy Instamart E-Commerce Database

2.0 Introduction

The entities required for the Swiggy Instamart E-Commerce Database are identified based on business requirements of the online grocery delivery platform. Each entity contains attributes with appropriate primary keys, foreign keys, unique keys and constraints. The relationships between entities define how customers, products, orders, payments and deliveries interact within the system. This analysis serves as the foundation for designing the ER Diagram, Relational Schema and database implementation while ensuring data integrity and consistency.

2.1 Entity Analysis

Entity: Customer

Attribute	**Key Type**
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	-
Email	Unique Key
Phone_Number	Unique Key
Password	Not Null
Address	Not Null
Pincode	-

Entity: Category

Attribute	**Key Type**
Category_ID	Primary Key
Category_Name	Unique Key

Description	-

Entity: Product

Attribute	**Key Type**
Product_ID	Primary Key
Category_ID	Foreign Key
Product_Name	Not Null
Brand	-
Price	Not Null
Stock	-
Weight	-
Expiry_Date	-
Image_URL	-

Entity: Cart

Attribute	**Key Type**
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_Items	-
Grand_Total	-

Entity: Cart_Item

Attribute	**Key Type**
CartItem_ID	Primary Key

Cart_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-

Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	-
Total_Amount	-

Entity: Order_Item

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Unit_Price	-

Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key

Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

Entity: Delivery

Attribute	**Key Type**
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_Partner	-
Delivery_Address	-
Delivery_Status	-
Delivery_Time	-

Entity: Review

Attribute	**Key Type**
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Review_Text	-
Review_Date	-

2.2 Entity Relationships

Entity 1	Relationship	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial/Total
Customer	Places	Order	One-to-Many	Partial/Total
Customer	Adds	Product (via Cart_Item)	Many-to-Many	Partial/Partial
Category	Contains	Product	One-to-Many	Total/Total
Cart	Contains	Cart_Item	One-to-Many	Total/Total
Order	Includes	Order_Item	One-to-Many	Total/Total
Order	Has	Payment	One-to-One	Total/Total
Order	Has	Delivery	One-to-One	Total/Total
Customer	Writes	Review	One-to-Many	Partial/Total
Product	Receives	Review	One-to-Many	Partial/Total

2.3 CONCLUSION

The Swiggy Instamart E-Commerce Database has been designed to efficiently manage the core operations of an online grocery delivery platform. The database includes essential entities such as Customer, Category, Product, Cart, Cart_Item, Order, Order_Item, Payment, Delivery, and Review, along with their attributes, keys, and relationships. These entities work together to ensure accurate data storage, maintain data integrity, and minimize redundancy.

The Entity Relationship (ER) model and relational schema provide a well-structured foundation for implementing the database. By using appropriate primary keys, foreign keys, unique constraints, and relationships, the system supports efficient order processing, inventory management, payment tracking, customer reviews, and delivery management.

Overall, this database design offers a scalable, reliable, and organized solution for the Swiggy Instamart platform. It can be further enhanced by integrating advanced features such as real-time inventory updates, AI-based product recommendations, personalized offers, and analytics dashboards to improve business performance and customer satisfaction.